David Batty

Autonomous Robotics Engineer | AI & Cyber-Physical Systems Architecture

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Profile

Robotics and cyber-physical systems engineer with architecture ownership across humanoid robotics, data pipelines and full website architecture for a launch-stage research institute. Completing a PhD in Electrical Engineering focused on radiation-aware navigation and planning for autonomous robotic exploration in unknown hazardous nuclear environments. Experience spans AI system development, FPGA edge-compute architecture, complex sensor fusion, ROS1/ROS2 autonomy, C++/Python development, CI/CD workflows, embedded hardware/firmware and defence/hazardous-environment R&D.

Architecture & Technical Fit

System architecture	Humanoid robot system architecture; data-pipeline ownership; FPGA edge-compute architecture; full ICMED website architecture; full-system coordination and operational standards.
Autonomy & robotics	ROS1/ROS2; Gazebo; wheeled, tracked and quadruped mobile robots; autonomous exploration; motion/path planning; occupancy-map abstraction; optical SLAM.
AI & sensing	AI system development; complex sensor fusion for advanced surroundings sensing; radiation-distribution prediction; VR biomarker-based immersion assessment.
Software & CI/CD	Python; C++ object-oriented programming; HTML; CI/CD workflows for humanoid robot and ICMED website development; full-stack platform development, deployment and maintenance; firmware.
High-consequence domains	Defence R&D; hazardous nuclear environments; conflict medicine; medical device evaluation; field-deployable technologies.

Professional Engineering Experience

AI & Cyber-Physical Systems Engineer *Aug 2025 – Present*

Virtual Engineering Centre, University of Liverpool *Liverpool, UK*

- Own humanoid robot system architecture and data pipelines, with full-system responsibility across software, sensing and integration activity.
- Coordinate colleagues from multidisciplinary teams to align interfaces, data flow and standards of operation across the humanoid robot platform.
- Work on FPGA edge-compute architecture within AI and cyber-physical systems engineering.
- Build CI/CD workflows for humanoid robot development, supporting repeatable integration and development practices.
- Develop AI systems and complex sensor-fusion approaches for advanced surroundings sensing; also develop systems that assess VR immersion using biomarkers and responses to in-game disruption.

Director of Engineering *2025 – Present*

Institute of Conflict Medicine (ICMED) *United Kingdom*

- Own full website architecture for a newly launching research institute with high-level clients, including CI/CD workflows for repeatable development, deployment and maintenance.
- Design, develop and deploy the institute's digital presence, establishing its online identity and communication infrastructure.
- Research and develop engineering solutions for conflict medicine, including medical device evaluation, field-deployable technologies and evidence-based methodologies that support clinical outcomes in conflict environments.

Post-Doctoral Research Associate *Jul 2025 – Sep 2025*

University of Liverpool *Liverpool, UK*

- Developed systems to assess VR immersion using biomarkers and evaluate responses to game disruption, supporting analysis of effects on player experience.

Electronic R&D Engineer *May 2017 – Dec 2019*

AWE *Aldermaston, UK*

- Developed a digital inertial navigation system using digital techniques to compare with analogue methodologies for trajectory tracking.
- Developed an environmental monitoring system, including hardware and firmware, for a sounding rocket test flight to monitor onboard transit and in-flight conditions.
- Investigated novel silicon-based 3D printing technology and protocols.

Training Admin and Systems Support *Aug 2023 – Sep 2025*

North West Medical Solutions *St Helens, UK*

- Developed, deployed and maintained a Python/HTML full-stack platform that streamlined coursework and paperwork workflows.
- Supported medical course delivery and operations, including course organisation and processing official paperwork for course awards.

Doctoral Robotics Research & Education

PhD, Electrical Engineering *Sept 2021 – Jul 2026*
University of Liverpool *Liverpool, UK*

- Dissertation: "Radiation Aware Navigation and Planning for Mobile Robotic Exploration of Unknown Hazardous Nuclear Environment."
- Developed motion and path-planning algorithms using occupancy-map abstraction and radiation-distribution prediction.
- Developed ROS1/ROS2 systems for wheeled, tracked and quadruped mobile robots for fully autonomous exploration, using simulation and hardware systems for validation.
- Collaborated with RACE, UKAEA and Oxford Robotics Institute on comparative studies against state-of-the-art solutions.

MEng, Electrical and Electronic Engineering *Graduated May 2018*
Heriot-Watt University *Edinburgh, UK*

- Completed modules in Electronics, Programming, Embedded Systems, Project Management, Engineering Mathematics and Power Systems.
- Dissertation: optical SLAM mapping system using budget hardware for remote semi-autonomous navigation of a mobile ground robot.

Teaching & Supervision

Lab Demonstrator, Department of Electronic and Electrical Engineering *Sept 2022 – Present*
University of Liverpool *Liverpool, UK*

- Delivered Robotics 1 and Robotics 2 undergraduate labs for cohorts of approximately 60 students per semester, covering C++ OOP, Gazebo, ROS1, ROS2, embedded robotics development and practical image processing.
- Supervised two DAAD German intern students during a three-month electronics hardware project placement.

Associate Tutor, Department of Computer Science *Sept 2025 – Dec 2025*
Edge Hill University *Ormskirk, UK*

- Assisted with delivery of the Foundations of Robotics and AI module.

Selected Publications, Presentations & Awards

- Batty D. et al., "Real-World Testing of Ultrasonic Beacons for Mobile Robot Radiation Emulation", Springer Lecture Notes in AI (TAROS 2024), published Dec 2024.
- Batty D. et al., "Occupancy Map Abstraction for Higher Level Mission Planning of Autonomous Robotic Exploration in Hazardous Nuclear Environments", Springer LNCS (TAROS 2023), published Oct 2023.
- Guest Speaker: "Using Robotics in Hazardous Environments", International Night of Science, Zaporizhzhia National Technical University, Sept 2025.
- Fukushima Decommissioning Topical Workshop: "Radiation Aware Navigation and Planning for Mobile Robot Exploration for Hazardous Environments", Oct 2024.
- UNTF 2024 oral presentation: "Occupancy Map abstraction for radiation aware navigation", Jul 2024; awarded runner-up.
- RAICo Junior Fellowship, 2023–2024: quadruped robot hardware testing and sensor development for real-world performance validation. GREEN CDT PhD Studentship, 2021–2025.

Memberships, Training & Operational Leadership

- **MIET - IET Membership**, 2025–present.
- NTHU Reactor and Detector Training Program, Taiwan, Feb 2022; Certificate in Nuclear Science and Engineering, University of Manchester, 2021.
- Qualified Emergency Medical First Responder and Beach Lifeguard; event medic and water-safety operator with event commander/controller experience leading teams of up to 25 personnel for events of up to 25,000 attendees.